3 Quantum statistical physics – single particles and waves

The problems are roughly in order of difficulty. Those indicated with ♡ are ones which are highly recommended. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

♡*Problem 3.1* **Internal energy and specific heat of an SHO**
In lectures we derived the free energy of the SHO (disregarding the zero point term):

$$F(T) = k_{\mathrm{B}} T \ln\left(1 - \mathrm{e}^{-\beta\hbar\omega}\right) \ .$$

Then:

$$\begin{aligned} U &= \frac{\partial}{\partial\beta}(\beta F) \\ &= \frac{\partial}{\partial\beta} \ln\left(1 - \mathrm{e}^{-\beta\hbar\omega}\right) \\ &= \frac{(-1)(-\hbar\omega)\mathrm{e}^{-\beta\hbar\omega}}{1 - \mathrm{e}^{-\beta\hbar\omega}} \\ &= \frac{\hbar\omega}{\mathrm{e}^{+\beta\hbar\omega} - 1} \ . \end{aligned}$$

For the specific heat:

$$\begin{aligned} C &= \frac{\mathrm{d}U}{\mathrm{d}T} \\ &= \frac{\mathrm{d}}{\mathrm{d}T} \frac{\hbar\omega}{\mathrm{e}^{+\beta\hbar\omega} - 1} \\ &= \frac{\hbar\omega}{\left(\mathrm{e}^{+\beta\hbar\omega} - 1\right)^2} (-1)\hbar\omega \ \mathrm{e}^{+\beta\hbar\omega} \underbrace{\frac{\partial\beta}{\partial T}}_{=-1/(k_{\mathrm{B}}T^2)} \\ &= k_{\mathrm{B}} \frac{\left(\hbar\omega/(k_{\mathrm{B}}T)\right)^2 \mathrm{e}^{+\beta\hbar\omega}}{\left(\mathrm{e}^{+\beta\hbar\omega} - 1\right)^2} \ . \end{aligned}$$

For high temperatures, the denominator is $\left(\hbar\omega/(k_{\mathrm{B}}T)\right)^2$, so $C \simeq k_{\mathrm{B}}$ as expected for one oscillator.

♡*Problem 3.2* **Spin waves in a ferromagnet**
Use the expression for the free energy for a collection of oscillators, derived in the context of photons, and disregard the zero-point term

$$F(T, V) = k_{\mathrm{B}} T \sum_{\mathbf{m}} \ln(1 - \mathrm{e}^{-\beta\hbar\omega_{\mathbf{m}}}) \ .$$

Then:

$$F_{\mathsf{sw}}(T,V) = \left(\frac{L}{2\pi}\right)^3 4\pi \int_0^{k_{\mathsf{m}}} k^2 \mathrm{d}k \; k_{\mathrm{B}} T \ln(1 - \mathrm{e}^{-\beta\hbar\omega_{\mathbf{k}}}) \; .$$

We may use the change of variable:

$$\omega = Ak^2 \quad \Rightarrow \quad k = \left(\frac{\omega}{A}\right)^{1/2} \quad \Rightarrow \quad \mathrm{d}k = \frac{1}{2}\frac{1}{(\omega A)^{1/2}}\mathrm{d}\omega \; .$$

So, with $\omega_{\mathsf{m}} = Ak_{\mathsf{m}}^2$:

$$\begin{aligned}
F_{\mathsf{sw}}(T,V) &= \left(\frac{L}{2\pi}\right)^3 2\pi \int_0^{\omega_{\mathsf{m}}} \frac{\omega}{A}\frac{1}{(\omega A)^{1/2}}\mathrm{d}\omega \; k_{\mathrm{B}} T \ln(1 - \mathrm{e}^{-\beta\hbar\omega}) \\
&= \left(\frac{L}{2\pi}\right)^3 2\pi \frac{k_{\mathrm{B}} T}{A^{3/2}} \int_0^{\omega_{\mathsf{m}}} \omega^{1/2}\mathrm{d}\omega \; \ln(1 - \mathrm{e}^{-\beta\hbar\omega}) \\
\text{(Let } x = \beta\hbar\omega.) \quad &= \left(\frac{L}{2\pi}\right)^3 2\pi \frac{(k_{\mathrm{B}} T)^{5/2}}{(\hbar A)^{3/2}} \int_0^{\beta\hbar\omega_{\mathsf{m}}} x^{1/2}\mathrm{d}x \; \ln(1 - \mathrm{e}^{-x})
\end{aligned}$$

At this point we have answered the question.

However at low temperatures, $\beta\hbar\omega_{\mathsf{m}} \gg 1$, we may harmlessly extend the region of integration to $\omega = \infty$, as the contribution is strongly suppressed. Then, integrating by parts and noticing that the contributions at the limits vanish:

$$\begin{aligned}
F_{\mathsf{sw}}(T,V) &\underset{T\to 0}{\sim} \left(\frac{L}{2\pi}\right)^3 2\pi \frac{(k_{\mathrm{B}} T)^{5/2}}{(\hbar A)^{3/2}} \int_0^{\infty} x^{1/2}\mathrm{d}x \; \ln(1 - \mathrm{e}^{-x}) \\
&= \left(\frac{L}{2\pi}\right)^3 2\pi \frac{(k_{\mathrm{B}} T)^{5/2}}{(\hbar A)^{3/2}} \left(\left[\frac{2}{3}x^{3/2}\ln(1 - \mathrm{e}^{-x})\right]_0^{\infty} - \frac{2}{3}\int_0^{\infty} \mathrm{d}x \; \frac{x^{3/2}}{1 - \mathrm{e}^{-x}}\right) \\
\text{(Use Baierlein p. 422.)} \quad &= -\left(\frac{L}{2\pi}\right)^3 2\pi \frac{(k_{\mathrm{B}} T)^{5/2}}{(\hbar A)^{3/2}} \frac{2}{3}\zeta(5/2)\Gamma(5/2) \\
&\simeq -1.006 \times \left(\frac{L}{2\pi}\right)^3 \frac{4\pi^{3/2}}{3}\frac{(k_{\mathrm{B}} T)^{5/2}}{(\hbar A)^{3/2}} \; .
\end{aligned}$$

Problem 3.3 **Cosmic Microwave Background and a photon gas**

We know that:

$$F_{\mathsf{opt}}(T,V) = -\frac{8\pi^5}{45}\frac{(k_{\mathrm{B}} T)^4}{\mathsf{h}^3 \mathrm{c}^3} V \; .$$

Thus:

$$S = -\left(\frac{\partial F}{\partial T}\right)_V = +k_{\mathrm{B}}\frac{32}{45}\pi^5 \left(\frac{k_{\mathrm{B}} T}{\mathsf{h}\mathrm{c}}\right)^3 V \tag{3.1}$$

$$\begin{aligned}
\text{(And:)} \quad U = \frac{\partial}{\partial\beta}(\beta F) &= -\frac{8\pi^5}{45}\frac{V}{\mathsf{h}^3\mathrm{c}^3}\frac{\partial}{\partial\beta}\frac{1}{\beta^3} \\
&= (-1)\frac{8\pi^5}{45}\frac{V}{\mathsf{h}^3\mathrm{c}^3}\frac{-3}{\beta^4} \\
&= \frac{8\pi^5}{15}\frac{(k_{\mathrm{B}} T)^4}{\mathsf{h}^3\mathrm{c}^3} V \\
&= 3pV \; .
\end{aligned}$$

Interesting to note that one can express the pressure for the ideal gas and photon gas in a similar form in terms of U:

$$p_{\text{opt}} = \frac{U}{3V} \quad \text{and} \quad p_{\text{gas}} = \frac{2}{3}\frac{U}{V} \ .$$

Since the photons were not interacting with matter, no heat left or entered the photon gas - and hence photon entropy, S_0, was conserved.

Thus:

$$S_0 = k_{\text{B}}\frac{32}{45}\pi^5\left(\frac{k_{\text{B}}T}{\text{hc}}\right)^3 V_{\text{univ}} \quad \Rightarrow \quad k_{\text{B}}T(V_{\text{univ}}) = \left(\frac{45}{32\pi^5}\frac{S_0}{k_{\text{B}}}\frac{(\text{hc})^3}{V_{\text{univ}}}\right)^{1/3} \ .$$

Notice the power of V – the linear dimensions of the "Universe" and the temperature are reciprocally related. so if the temperature at which photons decoupled from matter was, say, 3000 K and they are now at 3 K, then the "linear dimension" of the Universe was only a thousandth of its current extent.

Problem 3.4 **Contributions to the pressure within the Sun**
If we note that the contributions to the pressure from light, p_{opt}, and particles, p_{part}, vary very differently with respect to temperature:

$$p_{\text{opt}} = \frac{8\pi^5}{45}\frac{(k_{\text{B}}T)^4}{(\text{hc})^3} \quad \text{and} \quad p_{\text{part}} = \frac{N}{V}k_{\text{B}}T \ .$$

So one might expect the contribution from the photons to be larger in the centre of the Sun due to the fourth power of temperature.

$$\frac{p_{\text{opt}}}{p_{\text{part}}} = \frac{8\pi^5}{45}\frac{V}{N}\frac{(k_{\text{B}}T)^3}{(\text{hc})^3}$$

Now

$$\frac{k_{\text{B}}}{\text{hc}} \simeq \frac{1.4\times10^{-23}}{6.6\times10^{-34}\times3\times10^8} \simeq 70(\text{K m})^{-1} \quad \text{and} \quad \frac{8\pi^5}{45} \simeq 50 \ ,$$

So

$$\frac{8\pi^5}{45}\left(\frac{k_{\text{B}}}{\text{hc}}\right)^3 \simeq 50\times7^3\times10^3\,(\text{K m})^{-3} = 1.7\times10^7\,(\text{K m})^{-3} \ .$$

Thus

$$\frac{p_{\text{opt}}}{p_{\text{part}}} = 1.7\times10^7\,(\text{K m})^{-3}T^3\frac{V}{N} \ .$$

1. *The photosphere* We have $T_{\text{p-sphere}} = 5800\,\text{K}$ and $(V/N)_{\text{p-sphere}} = 10^{-23}\,\text{m}^{-3}$, so

$$\left.\frac{p_{\text{opt}}}{p_{\text{part}}}\right|_{\text{p-sphere}} = 1.7\times5.8^3\times10^{16}\times10^{-23} \simeq 3.3\times10^{-5} \ .$$

2. *The centre of the Sun* There, $T_{\text{cent}} = 1.5\times10^7\,\text{K}$ and the inverse particle number density $(V/N)_{\text{cent}} = 2\times10^{-32}\,\text{m}^{-3}$, so

$$\left.\frac{p_{\text{opt}}}{p_{\text{part}}}\right|_{\text{cent}} = 1.7\times1.5^3\times10^{28}\times2\times10^{-32} \simeq 1.1\times10^{-3} \ .$$

   Stars more massive than the Sun have higher central temperatures, so become radiation dominated in their centres.

Problem 3.5 **Dust in the Solar System**
The radiation pressure on a dust grain depends on its cross sectional area, πa^2, (where a is the radius of the grain) presented to the Sun. In detail this depend on absorption versus reflection of radiation (and emissivity) and factors of cosines etc. We will be crude and assume that we may use the radiation pressure of the photons from the Sun's photosphere streaming towards the Earth. The repulsive force on the grain due to radiation pressure for a grain at radial distance R from the Sun is ($R_\odot$ is the radius of the Sun)

$$F_{\text{opt}}^{\text{grain}} = \pi a^2 \left(\frac{R_\odot}{R}\right)^2 \frac{8\pi^5}{45} \frac{(k_{\text{B}} T_{\text{p-sphere}})^4}{(\text{hc})^3} \ .$$

The attractive force due to the Sun's gravitational field is, where ρ is the mass density of the grain and $M_\odot$ is the Sun's mass:

$$F^{\text{grav}} = -\frac{GM_\odot}{R^2} \underbrace{\frac{4\pi}{3}\rho a^3}_{\text{mass grain}} \ .$$

Because both effects are inversely proportional to R^2, the distance from the Sun is immaterial. But the value of a is not. Equating the two expressions we see that the effects balance with a grain radius of

$$a_{\text{balance}} = \frac{2\pi^4}{15} \frac{R_\odot^2 (k_{\text{B}} T_{\text{p-sphere}})^4}{GM_\odot \rho (\text{hc})^3} \ .$$

For $a > a_{\text{balance}}$, the grains will fall towards the Sun, and will flow away from the Sun if $a < a_{\text{balance}}$, due to the balance of surface to volume effects.

♡*Problem 3.6* **Adiabatic expansion of a photon gas**
We wish to find the counterpart to $pV^\gamma =$ constant for an adiabatic process for the photon gas.

For an adiabatic process the entropy is constant, say S_0. We use Eq. (3.1), i.e.

$$\frac{S_0}{k_{\text{B}}} = \frac{32}{45}\pi^5 \left(\frac{k_{\text{B}} T}{\text{hc}}\right)^3 V \tag{3.2}$$

We know the radiation pressure:

$$\begin{aligned} p &= \frac{8\pi^5}{45} \frac{(k_{\text{B}} T)^4}{(\text{hc})^3} \\ \Rightarrow \quad \frac{p}{\text{hc}} \frac{45}{8\pi^5} &= \left(\frac{k_{\text{B}} T}{\text{hc}}\right)^4 \\ \Rightarrow \quad \left(\frac{p}{\text{hc}} \frac{45}{8\pi^5}\right)^{3/4} &= \left(\frac{k_{\text{B}} T}{\text{hc}}\right)^3 \ . \end{aligned} \tag{3.3}$$

We may eliminate temperature by substituting this into Eq. (3.2), we see

$$\begin{aligned}
\frac{S_0}{k_{\mathrm{B}}} &= \frac{32}{45}\pi^5 \left(\frac{p}{\mathsf{hc}}\frac{45}{8\pi^5}\right)^{3/4} V \\
\Rightarrow \quad \frac{1}{4}\left(\frac{45}{32\pi^5}\right)^{1/4} \frac{S_0}{k_{\mathrm{B}}} &= \left(\frac{p}{\mathsf{hc}}\right)^{3/4} V \\
\Rightarrow \quad \mathsf{hc}\left(\frac{1}{4^4}\frac{45}{32\pi^5}\right)^{1/3} \left(\frac{S_0}{k_{\mathrm{B}}}\right)^{4/3} &= pV^{4/3} \ .
\end{aligned}$$

This is the counterpart of $pV^\gamma =$ constant.

Problem 3.7 **Effective, or average, speed of sound.**
We generalise from Eqn (7.4), which is

$$F_{\mathsf{opt}}(T,V) = -\frac{4\pi^5}{45}\frac{(k_{\mathrm{B}}T)^4}{\mathsf{h}^3}V\ \frac{2}{\mathrm{c}^3}\ ,$$

where we have isolated the factor of 2 for polarisations and the reciprocal cube of the velocity of light.

For elastic waves in a solid with two transverse modes (or $d-1$ in d-dimensions) and one longitudinal mode (there is always one independent of dimension), we have speeds of sound c_{t} and c_ℓ respectively. (We will relax the assumption of the two transverse branches having the same velocity at end.)

Then addiing the free energies associated with these modes (as we added the two polarisations before), we see that

$$\begin{aligned}
F_{\mathsf{elas}} &= -\frac{4\pi^5}{45}\frac{(k_{\mathrm{B}}T)^4}{\mathsf{h}^3}V\ \left[\frac{1}{\mathrm{c}_\ell^3}+\frac{d-1}{\mathrm{c}_{\mathsf{t}}^3}\right] \\
&= -\frac{4\pi^5}{45}\frac{(k_{\mathrm{B}}T)^4}{\mathsf{h}^3}V\ \left[\frac{\mathrm{c}_{\mathsf{t}}^3+(d-1)\mathrm{c}_\ell^3}{\mathrm{c}_{\mathsf{t}}^3\mathrm{c}_\ell^3}\right] \\
\text{(Define average velocity, } \bar{\mathrm{c}}\text{:)} \quad &= -\frac{4\pi^5}{45}\frac{(k_{\mathrm{B}}T)^4}{\mathsf{h}^3}V\frac{d}{\bar{\mathrm{c}}^3}
\end{aligned}$$

So we see that

$$\bar{\mathrm{c}} = \frac{d^{1/3}\ \mathrm{c}_\ell\mathrm{c}_{\mathsf{t}}}{(\mathrm{c}_{\mathsf{t}}^3+(d-1)\mathrm{c}_\ell^3)^{1/3}}\ .$$

If all the transverse speeds of sound are different, $\mathrm{c}_{\mathsf{t},i}$, with $i=1,2,\cdots,(d-1)$, then we have:

$$\frac{d}{\bar{\mathrm{c}}^3} = \frac{1}{\mathrm{c}_\ell^3} + \sum_{i=1}^{(d-1)}\frac{1}{\mathrm{c}_{\mathsf{t},i}^3}\ .$$

This is not readily simplified.